The Standard Model Gauge Group from $J_3(\mathbb{O})$: Color, Weak Isospin, and Hypercharge from the Mediator and Its Idempotent

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Abstract. Todorov and Dubois-Violette showed the Standard Model gauge group is the intersection of two maximal subgroups of $F_4 = \mathrm{Aut}(J_3(\mathbb{O}))$; Boyle gave the idempotent-stabilizer form of the construction; Baez and Schwahn generalized it to any suitable pair of Jordan subalgebras. We show this project's own octonionic construction — built independently, without reference to that program — reproduces it exactly, using only two objects already load-bearing throughout this corpus: the mediator direction fixing $\mathrm{Im}(\mathbb{O})$'s complex structure, and a single idempotent it determines inside $J_3(\mathbb{O})$. The mediator's own stabilizer in $\mathfrak{g}_2 = \mathrm{Der}(\mathbb{O})$, embedded in $\mathfrak{f}_4 = \mathrm{Der}(J_3(\mathbb{O}))$, is verified identical in construction to the published program's color $\mathfrak{su}(3)$. Its centralizer in $\mathfrak{f}_4$ is a second, independent $\mathfrak{su}(3)$; adjoining $L(J_0)$, the traceless Jordan-multiplication operators, to all of $\mathfrak{f}_4$ generates $\mathfrak{e}_6$, verified of the noncompact real form $\mathfrak{e}_{6(-26)}$ by direct Killing-form diagonalization, with the second $\mathfrak{su}(3)$ sitting inside it. The mediator's own idempotent breaks this second $\mathfrak{su}(3)$ to $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$, verified exactly 4-dimensional with a 1-dimensional center. Three branching theorems follow. One — the $8/1/7$ count, the d_R-type singlet ratio of exactly -2, and the genuine absence of a u_R-type ratio-$+4$ state — was derived from the abstract branching rule and stated in advance of the computation that confirmed it, exactly. The other two were found by direct computation first and proven afterward: the lepton-to-quark doublet hypercharge ratio, measured at exactly 3, matches the Standard Model's own $(-\frac{1}{2})/(\frac{1}{6}) = -3$ and is then shown to be the forced $\mathfrak{su}(3) \supset \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ adjoint-to-fundamental ratio; Peirce grade, discovered by direct subspace comparison to coincide exactly with weak-doublet membership, is then shown identical to it by an exact index-counting argument. Relative to this specific idempotent, the full matter content — 8 doublets, 1 triplet, 7 singlets, 26 states total — is exhaustive: no further generation-shaped content hides in another Peirce grade, because Peirce grade and weak type are provably the same partition. This is consistent with, not contrary to, the published program's own stated boundary: a full chiral generation needs the complexification $J_3(\mathbb{O}) \otimes \mathbb{C}$ or the bi-Cayley extension, not $J_3(\mathbb{O})$ alone. The corpus's own distinctive contribution is named precisely: a derivation of the octonionic seed's *selection* — via this project's separately-established enumeration theorem — where the published program assumes octonions as a starting choice.

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1. Introduction

Dubois-Violette and Todorov's own discovery — the Standard Model gauge group as the intersection of two maximal subgroups of $F_4 = \mathrm{Aut}(J_3(\mathbb{O}))$ — has grown, in Boyle's explicit idempotent-stabilizer form and Baez–Schwahn's generalization to arbitrary suitable Jordan subalgebra pairs, into an active research program building the Standard Model's structure directly from the exceptional Jordan algebra. This note reports an independent arrival at the same building, from a route this program does not use: not a chosen octonion subalgebra pair, but the mediator direction and idempotent this corpus's own earlier work already made load-bearing, for entirely different reasons, well before any gauge-structure question was posed.

What follows is a full account of a chain built and checked stage by stage, each stage verified directly against the raw octonion multiplication table and $J_3(\mathbb{O})$'s own Jordan product — no step assumed from citation where direct computation was possible, matching the standard this corpus's companion papers already set for $J_3(\mathbb{O})$ and $\mathfrak{f}_4$ (Ref. [companion, §§2–4]) and the TKK/$\mathfrak{e}_7$ construction (Ref. [companion, TKK]).

The chirality boundary, stated once, prominently, rather than left implicit. Everything in this paper lives in the *real* $J_3(\mathbb{O})$. The published program's own stated wall — that F_4 and real $J_3(\mathbb{O})$ carry only real or pseudo-real representations, and a genuinely chiral gauge theory needs complexification — applies here identically, and is not evaded by anything in this construction. What is shown is the gauge group and the matter representation's quantum numbers *up to conjugation*; no left/right distinction is available, or claimed, anywhere below.

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2. Color: the mediator's $\mathfrak{su}(3)$

[FACT, proven, companion paper] $\mathfrak{g}_2 = \mathrm{Der}(\mathbb{O})$ is 14-dimensional, built directly from the raw octonion multiplication table as the null space of the derivation-property linear system, verified to machine precision.

[FACT, proven] G_2 acts transitively on the unit sphere in $\mathrm{Im}(\mathbb{O})$; the mediator direction e, already load-bearing throughout this corpus for the geometric skeleton (σ, J, g), is an orbit coordinate, not a further commitment. Its stabilizer in $\mathfrak{g}_2$ is exactly 8-dimensional, verified directly (the null space of $\{D \in \mathfrak{g}_2 : D(e) = 0\}$), and its action on the six non-mediator directions of $\mathrm{Im}(\mathbb{O})$ is the genuine fundamental representation of $\mathfrak{su}(3)$: traceless, anti-Hermitian, and irreducible (Schur's lemma, commutant dimension exactly 1), all verified directly against the explicit 3×3 complex matrices obtained via the mediator's own complex structure $J = L_e$.

[FACT, proven] This $\mathfrak{su}(3)$ embeds in $\mathfrak{f}_4 = \mathrm{Der}(J_3(\mathbb{O}))$ via the natural octonion-entry-wise extension — each generator acting identically on all three off-diagonal octonion blocks of $J_3(\mathbb{O})$, diagonal entries fixed — verified to satisfy the derivation property on $J_3(\mathbb{O})$'s Jordan product to machine precision.

This is, by construction, identical to the published program's own color $\mathfrak{su}(3)$: Todorov–Dubois-Violette's color group is the stabilizer, within G_2, of a fixed unit imaginary octonion — the same construction, arrived at independently.

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3. $\mathfrak{e}_6$: the second $\mathfrak{su}(3)$ and the noncompact real form

[FACT, proven] $J_3(\mathbb{O})$ is built directly (elements (a_1, a_2, a_3, a, b, c), $a_i \in \mathbb{R}$, $a, b, c \in \mathbb{O}$, Jordan product $X \circ Y = \frac{1}{2}(XY + YX)$ via ordinary matrix multiplication with octonion entries) and verified to satisfy the defining Jordan identity to machine precision, alongside commutativity and correct identity-element behavior.

[FACT, proven] $\mathfrak{f}_4 = \mathrm{Der}(J_3(\mathbb{O}))$ is built directly, as the null space of a 10206×729 derivation-constraint system spanning all $\binom{27}{2}$ basis-element pairs, and verified exactly 52-dimensional.

[FACT, proven] The centralizer of the mediator's $\mathfrak{su}(3)$ within $\mathfrak{f}_4$ is exactly 8-dimensional, closed under the bracket (verified, residuals at machine precision), and semisimple (Killing-form eigenvalues uniformly negative, no zero eigenvalues) — a second, independent $\mathfrak{su}(3)$.

[FACT, proven] $L(J_0)$, the Jordan-multiplication operators of the 26-dimensional traceless subalgebra, is 26-dimensional and satisfies the standard reduced-structure-algebra closure by two classical identities, each stated and directly verified rather than left to sampling alone: $[D, L_x] = L_{D(x)}$ for any derivation D and any $x \in J_0$ — immediate from the derivation property applied to $D(x \circ y) = D(x) \circ y + x \circ D(y)$, confirmed to machine precision — giving $[\mathfrak{f}_4, L(J_0)] \subseteq L(J_0)$ outright, not merely observed across sampled pairs; and $[L_x, L_y] \in \mathrm{Der}(J)$ for any x, y — the classical inner-derivation fact for Jordan algebras, confirmed directly against the derivation property to machine precision — giving $[L(J_0), L(J_0)] \subseteq \mathfrak{f}_4$ by the same standard, with zero leakage back into $L(J_0)$ (checked across 300 sampled pairs as confirmation on top of the identity, not in place of it) — the exact Cartan-decomposition signature of a symmetric pair.

[FACT, proven] $\mathfrak{e}_6 := \mathfrak{f}_4 \oplus L(J_0)$ is verified genuinely 78-dimensional (all generators linearly independent) and its Killing form, computed directly from its own structure constants and diagonalized, has signature exactly -26: 26 positive eigenvalues (on $L(J_0)$) and 52 negative (on $\mathfrak{f}_4$). This is $\mathfrak{e}_{6(-26)}$, the noncompact real form with maximal compact subalgebra $\mathfrak{f}_4$ — computed, not assumed.

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4. The idempotent A: electroweak breaking

[DEF] $A := (0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}e, 0, 0)$ — built from the same mediator direction e as §2's color $\mathfrak{su}(3)$, no independent choice.

[FACT, proven] $A \circ A = A$, verified to machine precision: a genuine idempotent.

[FACT, proven] L_A's spectrum on the full 27-dimensional $J_3(\mathbb{O})$, computed via the trace-form-corrected (genuinely self-adjoint) operator, is exactly $\{0, \frac{1}{2}, 1\}$ with multiplicities $\{10, 16, 1\}$ — the standard Peirce profile for a primitive idempotent ($J_1 = \mathbb{R}A$, $J_{1/2} \cong \mathbb{O}^2$, $J_0 \cong \mathfrak{h}_2(\mathbb{O})$), matching the companion paper's independently-verified profile for a diagonal idempotent exactly.

[FACT, proven] The stabilizer of A within the second $\mathfrak{su}(3)$ (the subalgebra commuting with L_A) is exactly 4-dimensional, closed under the bracket, verified to machine precision using the second $\mathfrak{su}(3)$'s own Killing form as the canonical metric (§7).

[FACT, proven] This 4-dimensional stabilizer splits exactly as a 1-dimensional center ($\mathfrak{u}(1)_Y$, verified commuting with all three remaining generators to machine precision) and a 3-dimensional, genuinely non-abelian complement ($\mathfrak{su}(2)_{\text{weak}}$, closure verified to machine precision via the Killing-form-orthogonal projection, §7).

[FACT, proven] $\mathfrak{u}(1)_Y$ commutes with the mediator's color $\mathfrak{su}(3)$ exactly (machine precision) — hypercharge is color-blind, as required.

A first, disqualified candidate, reported for completeness. The centralizer of the *full* mediator $\mathfrak{g}_2$ within $\mathfrak{f}_4$ is also exactly 3-dimensional and non-abelian — but its spin content on $J_3(\mathbb{O})$'s traceless 26 is exclusively triplets and quintets (Casimir ratio exactly 3, matching spin-2:spin-1), never doublets or singlets. An $\mathfrak{su}(2)$ acting on every matter component as vector or higher cannot be weak isospin; this is the structural reason the joint centralizer of $\mathfrak{su}(3)\oplus$(this $\mathfrak{su}(2)$) in $\mathfrak{e}_6$ vanishes exactly (verified: rank 78 of 78) despite $\mathfrak{e}_6$'s rank-6 room — not because $\mathfrak{e}_6$ lacks capacity, but because this specific $\mathfrak{su}(2)$ is disqualified by its own representation content before the question of room arises.

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5. Three branching theorems

Theorem 1 (the ratio-3 theorem). Under $\mathfrak{su}(3) \supset \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with $Y = \mathrm{diag}(a, a, -2a)$: the fundamental **3** splits as a doublet at charge a plus a singlet at charge $-2a$; the adjoint **8** splits as a triplet-and-singlet at charge 0 plus two doublets at charge $\pm 3a$ — the off-diagonal blocks linking the doublet rows to the third row, charge difference $a - (-2a) = 3a$. Any state census built from this breaking must show adjoint-doublet hypercharge exactly three times fundamental-doublet hypercharge, with no free parameter.

[FACT, verified] Checked directly: the color-singlet doublets (living in the second $\mathfrak{su}(3)$'s adjoint, i.e. $(1, 8)$ under color×electroweak) carry $Y = \pm 0.343$; the colored doublets (the fundamental pieces, $(3, 3) \oplus (\bar{3}, \bar{3})$) carry $Y = \mp 0.1143$. Ratio: $0.343/0.1143 = 3.00$ exactly. The Standard Model's own lepton-to-quark doublet ratio is $(-\frac{1}{2})/(\frac{1}{6}) = -3$ — the same Lie-theoretic number, reproduced from the branching alone, with zero free parameters between prediction and measurement.

Theorem 2 (the $8/1/7$ branching, with a pre-registered singlet prediction). The full traceless 26 branches under color×(second $\mathfrak{su}(3)$, pre-breaking) as $26 = (1, 8) \oplus (3, 3) \oplus (\bar{3}, \bar{3})$ — dimension check $8 + 9 + 9 = 26$. Breaking the second factor by Theorem 1's rule and collecting multiplicities forces exactly 8 doublets, 1 triplet, 7 singlets ($8 \times 2 + 1 \times 3 + 7 \times 1 = 26$), and predicts, before computation, that the seven singlets split as one color-singlet at $Y = 0$ (the adjoint's own neutral state) plus $3 \oplus \bar{3}$ colored singlets at Y-ratio -2 relative to the colored doublet (the fundamental's own $-2a$ against its a) — with a state at ratio $+4$ genuinely forbidden.

[FACT, verified] All predictions confirmed by direct computation, in advance of any adjustment: color Casimir on the 26 gives exactly 8 singlets (0) and $18 = 6 \times 3$ non-singlet states; weak-Casimir slicing gives exactly 16 doublet states, 3 triplet states, 7 singlet states; the singlet block's colored piece sits at $Y = \mp 0.2287$, ratio -2.001 against the colored doublet's 0.1143, on all six states; the color-singlet piece sits at $Y = 0$ exactly (after correcting a degenerate-subspace basis artifact, §7); no state appears anywhere at ratio $+4$.

Theorem 3 (Peirce grade is weak type, exactly). By F_4-transitivity on primitive idempotents of $J_3(\mathbb{O})$, and since the claim below is conjugation-invariant, take $A = E_{33}$ (the diagonal rank-one projection) without loss of generality — not §4's own mediator-tied A, which is F_4-conjugate to it. Peirce grade counts how many of a state's two matrix indices equal 3: none (Peirce-0, the $\mathfrak{h}_2(\mathbb{O})$ block on indices $\{1, 2\}$), one (Peirce-$\frac{1}{2}$, the two off-diagonal slots touching index 3), both (Peirce-1, the A-line itself). The weak $\mathfrak{su}(2)$ is exactly the stabilizer's rotation of indices $\{1, 2\}$. A state transforms as a doublet precisely when it carries exactly one index in $\{1, 2\}$ — verbatim the Peirce-$\frac{1}{2}$ condition. The two labels were never independent, because the idempotent that grades Peirce is the same object that breaks the weak group.

[FACT, verified] Checked three independent ways: the weak-doublet sector (all states with weak Casimir matching spin-$\frac{1}{2}$) and the Peirce-$\frac{1}{2}$ eigenspace of L_A, both extracted independently via distinct computations, span *the identical* 16-dimensional subspace of the 26 (rank of each alone, 16; rank of the union, 16 — not merely equal dimension, the same subspace). Peirce-0 (once corrected for a 1-dimensional trace-direction contamination not part of the traceless 26, §7) contains exactly the non-doublet remainder: 1 triplet $+7$ singlets, with representation content and Y-values identical to

Theorem 2's own findings, not new information arrived at a second way.

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6. The census: full color$\times$weak$\times Y$ table

Every value below was computed jointly-diagonalizing color's own Cartan subalgebra, the weak Cartan, and $\mathfrak{u}(1)_Y$ simultaneously (verified mutually commuting to machine precision throughout), with all degenerate-subspace extractions performed using the algebra's own canonical (Killing- or trace-form) metric rather than an arbitrary coordinate basis — see §7 for why this precaution was necessary and what it caught.

Piece	Color	Weak	States	Y (raw)	$Y/Y_{\text{colored doublet}}$
Lepton-type doublet	**1**	**2**	4 (incl. conj.)	± 0.343	± 3
Quark-type doublet	$\mathbf{3} \oplus \bar{\mathbf{3}}$	**2**	12 (incl. conj.)	∓ 0.1143	∓ 1
Sterile singlet	**1**	**1**	1	0	0
d_R-type singlet	$\mathbf{3} \oplus \bar{\mathbf{3}}$	**1**	6 (incl. conj.)	∓ 0.2287	∓ 2
Neutral triplet	**1**	**3**	3	0	0

Total: $4 + 12 + 1 + 6 + 3 = 26$, exact.

On normalization: the raw Y values are basis-scaled and individually meaningless; only ratios are physical. Fixing the colored-doublet value to the Standard Model's own $\frac{1}{6}$ reproduces the doublet and singlet rows exactly against the corresponding Standard Model fermion hypercharges, up to the conjugate-doubling explained below. The neutral triplet has no Standard Model fermion counterpart to compare against — its natural analogue is a gauge-adjoint slot, not a matter multiplet, and it is reported here as part of $J_3(\mathbb{O})$'s own content rather than checked against an external number.

On the doubling: every state count above is twice the naive Standard Model count. $J_3(\mathbb{O})$ is real throughout; a complex representation embedded in a real space is necessarily accompanied by its own conjugate. One lepton doublet and its mirror, not two lepton doublets — the same reading applies to every colored entry.

On the u_R-type slot: absent, as Theorem 2 predicted. This is not a gap in the construction; it is $J_3(\mathbb{O})$'s own boundary, consistent with the published program's statement that a full generation needs the bi-Cayley or complexified extension (§9).

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7. Canonical metrics, a closed-form artifact, and two $U(1)$s

Method. Every computation above involving a degenerate eigenspace was interrogated using the algebra's own canonical metric — the Killing form for Lie-subalgebra complements, the trace form for $J_3(\mathbb{O})$'s own coordinate bases — and Hermitian-specific eigensolvers throughout. This is stated as a method because it is one: an arbitrary coordinate basis on a degenerate subspace is not canonically related to that subspace's own algebraic structure, and every place this corpus's own working notes recorded a spurious numerical residual traced to exactly this substitution, resolved exactly once the canonical metric replaced it.

A diagnosed artifact, not a numerical inconvenience. L_{A_0}'s spectrum on the full 27-dimensional $J_3(\mathbb{O})$ is exactly $\{-\frac{1}{3}, \frac{1}{6}, \frac{2}{3}\}$ with multiplicities $\{10, 16, 1\}$. Restricting to the traceless 26 via least-squares against a generic basis produces a spurious third eigenvalue, ≈ 0.333, in addition to the expected $-\frac{1}{3}$ and $\frac{1}{6}$. This is not basis noise: L_{A_0}, a Jordan-multiplication operator, does not preserve tracelessness — its non-derivation, $\mathfrak{p}$-side character (§3) showing up concretely. The single non-invariant direction is $v = E - 2A$, where $E = 1 - A$ is the identity of the Peirce-0 subalgebra ($E \circ E = E$, verified); v is traceless (verified, trace $= 0$ exactly) but $L_{A_0}(v)$ is not (verified, trace $= -2$ exactly), and the Rayleigh quotient $\langle v, L_{A_0} v\rangle / \langle v, v\rangle$, computed via the trace form, equals $2/6 = \frac{1}{3}$ exactly — the observed artifact, to the digit, diagnosed rather than merely bounded.

Two $U(1)$s, a containment rather than an identity. The idempotent A generates two distinct $U(1)$s at two depths of its own stabilizer chain: a coarse one on the $\mathfrak{p}$-side, L_{A_0}'s own real spectrum, genuinely compact once complexified but carrying only the Peirce-grade bit Theorem 3 already characterizes fully; and a fine one on the $\mathfrak{k}$-side, $\mathfrak{u}(1)_Y$ itself, the seven-value system carrying every branching theorem in §5. The coarse charge is constant on each of the fine structure's weak-type sectors, since both descend from the same A — a containment, not a coincidence, and not two names for one object.

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8. Discussion: an independent arrival, and what is genuinely new

The published program. Dubois-Violette and Todorov first showed G_{SM} is the intersection of two maximal subgroups of F_4; Boyle's explicit form — automorphisms preserving an idempotent Π, with $\{U, v, \varphi\} \in SU(3) \times SU(2) \times U(1)$ block-diagonal — is the direct model for §4's construction; Baez and Schwahn generalize the intersection to any Jordan subalgebra pair $A \cong \mathfrak{h}_2(\mathbb{O})$, $B \cong \mathfrak{h}_3(\mathbb{C})$, $A \cap B \cong \mathfrak{h}_2(\mathbb{C})$, with $\mathrm{Stab}(A) \cap \mathrm{Stab}(B)_0 \cong S(U(2) \times U(3))$. Krasnov's $\mathrm{Spin}(9)$ characterization sits alongside. This construction was arrived at independently of that literature, using objects — the mediator, its idempotent — already load-bearing in this corpus for unrelated reasons, well before any gauge question was posed; the coincidence in destination, not the route, is what this note reports.

What is genuinely this corpus's own. Not the destination — the published program reaches the same gauge group and the same branching structure by its own routes. What is distinctive: (i) a derivation, not an assumption, of *why* the octonions specifically — this project's separately-established enumeration theorem proves the $1{+}3{+}3$ non-associative grading is forced, not chosen, once non-associativity retention is demanded at all; the published program starts from $\mathbb{O}$ as given. (ii) The mediator as a single shared origin for both complexification (the J underlying this corpus's own quantization) and color (§2) — the same direction doing two jobs the published program treats as unconnected inputs. (iii) The full computational verification chain itself, every stage checked against raw multiplication tables rather than cited, in the discipline this corpus's companion papers already established for $J_3(\mathbb{O})$ and $\mathfrak{f}_4$ (Ref. [companion, §§2–4]).

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9. Scope

The chirality wall, stated at its published size. Everything above lives in real $J_3(\mathbb{O})$, which carries only real and pseudo-real representations. The census of §6 matches one Standard Model generation's quantum numbers *up to conjugation* — there is no left/right distinction available in this construction, and none is claimed. This is not a gap specific to this route; it is the published program's own known wall, and the reason its own complexification and bi-Cayley extensions exist. Nothing here closes that door; the mediator's own J (already load-bearing throughout this corpus for quantization) is a native candidate supplier for it, not yet verified to do that job.

What was verified exhaustive, and what was not. Theorem 3 establishes that no further generation-shaped content hides in another Peirce grade of $J_3(\mathbb{O})$ *relative to this specific* A; it does not establish that no other idempotent, or no extension beyond $J_3(\mathbb{O})$, could yield more. The published program's own placement of a full generation in the bi-Cayley triple rather than $J_3(\mathbb{O})$ alone is consistent with, not contradicted by, this exhaustion result.

Independent verification, owed and not yet discharged. This corpus's own two-implementation standard, applied throughout its companion papers, has not yet been applied here: the census numbers in §6 rest on one construction of $J_3(\mathbb{O})$, $\mathfrak{f}_4$, and the stabilizer chain. An independent rebuild, checked against these same numbers before they are treated as fully closed, remains outstanding.

What was not attempted. Gauge dynamics, a Lagrangian, the Higgs mechanism, or any claim beyond the group-theoretic and representation-theoretic content of §§2–6. This note follows this corpus's own house discipline: a statement of what is proven, at its correct weight, and nothing claimed beyond it.

References

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Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.